

Finite-sample density reconstruction from a single causal order

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<https://github.com/Quantlyra/quantyra-null-cone-lean>

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Abstract

We observe only the isomorphism class of the strict coordinatewise order of n independent points in the unit square. Their unknown density is smooth, lies between $1/2$ and $3/2$, has uniform marginals, and is Euclidean Lipschitz with constant at most two. We construct one measurable order-only estimator whose full-square supremum error, modulo one global coordinate exchange, is at most $\min\{1/2, 650(\log n/n)^{1/4}\}$ with probability at least $19/20$, uniformly over this class and for every $n \geq 2$. The principal reduction uses observed predecessor and successor counts: every retained point has reliable reconstructed coordinates, while every point in a sufficiently deep cell is retained. This removes a global discarded-mass penalty from local density estimation. We also obtain a finite inverse bound for the actual order laws. Smooth marginal-preserving Gaussian alternatives give both a finite two-point obstruction and a matching logarithmic lower bound: for $n \geq 2^{64}$, every eligible estimator, including independent randomization, has strict error greater than $(\log n/n)^{1/4}/8192$ with probability at least $1/2$ under some density in the same class. Thus the uniform 95% minimax radius has rate $(\log n/n)^{1/4}$ up to constants. The upper, law, lower and minimax-radius endpoints are Lean-certified. The lower argument already holds with coordinate data. Efficient computation, sharp constants and useful practical calibration remain open.

1 Experiment, results and contribution

Let $D = [0, 1]^2$. The class $\mathcal{K}$ consists of functions smooth on an open neighborhood of D satisfying, for $x, y \in D$ and $u, v \in [0, 1]$,

$$\begin{aligned} \frac{1}{2} \leq \rho(x) \leq \frac{3}{2}, \quad |\rho(x) - \rho(y)| \leq 2|x - y|_2, \\ \int_0^1 \rho(u, t) dt = 1, \quad \int_0^1 \rho(t, v) dt = 1. \end{aligned} \tag{1}$$

The norm in the Lipschitz condition is Euclidean. In particular $Q_\rho(du dv) = \rho(u, v) du dv$ on D is a probability measure. Every density is smooth, but no common bound on higher derivatives is assumed. Uniform marginals fix the coordinate gauge.

From $Z_i = (U_i, V_i)$, $1 \leq i \leq n$, drawn independently from Q_ρ , we observe the directed relation

$$i \prec j \iff U_i < U_j \text{ and } V_i < V_j$$

only up to vertex relabeling. Write $\mathcal{O}_n$ for this finite observation space and π_n^ρ for its actual pushforward law. One may include all directed Boolean relation codes in $\mathcal{O}_n$ and give invalid codes

a fixed fallback; this agrees with the formal implementation. No coordinate order, rank, embedding or extra sample is observed. The direction is retained: time reversal is not quotiented.

Let $T(u, v) = (v, u)$ and $G = \{\text{id}, T\}$. For bounded functions put

$$L(b, \rho) = \min_{S \in G} \|b - \rho \circ S\|_{\infty, D}, \quad d_{\text{conf}}(\rho, \sigma) = L(\rho, \sigma).$$

All norms are over the full closed square. One S must work at every point. More generally, success at radius a means

$$\exists S \in G \quad \forall x \in D, \quad |b(x) - \rho(Sx)| \leq a. \quad (2)$$

For unbounded outputs we use this definition directly (or an extended supremum), avoiding any boundedness assumption in the lower theorem. Total variation is $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$.

Theorem 1.1 (One-order upper guarantee). *For every integer $n \geq 2$ there exists a measurable estimator $\hat{\rho}_n : \mathcal{O}_n \rightarrow \{b : D \rightarrow [1/2, 3/2]\}$, independent of ρ , whose outputs are measurable functions, such that for every $\rho \in \mathcal{K}$,*

$$\pi_n^\rho \{L(\hat{\rho}_n, \rho) \leq R_n\} \geq \frac{19}{20}, \quad R_n = \min \left\{ \frac{1}{2}, 650 \left(\frac{\log n}{n} \right)^{1/4} \right\}. \quad (3)$$

The finite observation space makes the estimator and its deterministic success events measurable. The construction below also establishes measurability in the query point. Outputs need not be smooth, integrate to one or have uniform marginals; *density estimator* refers to the target being estimated, not membership of the output in $\mathcal{K}$.

Corollary 1.2 (Actual-law inverse). *For $\rho, \sigma \in \mathcal{K}$ and $n \geq 2$,*

$$d_{\text{conf}}(\rho, \sigma) \leq \min \left\{ 1, 1300 \left(\frac{\log n}{n} \right)^{1/4} + \frac{10}{9} \text{TV}(\pi_n^\rho, \pi_n^\sigma) \right\}. \quad (4)$$

This compares two probability laws; it is not an estimator of their total variation from a single observed order.

Theorem 1.3 (Finite randomized lower bound). *For $n \geq 1$ set*

$$h = \min\{1/2, n^{-1/4}\}, \quad f(t) = te^{-t^2}, \quad a_h(x) = f((x - 1/2)/h),$$

and $\rho_0 = 1$, $\rho_1(u, v) = 1 + 2ha_h(u)a_h(v)$. These two densities belong to $\mathcal{K}$. For any estimator from $\mathcal{O}_n$, including an arbitrary independent probability-space random seed, assume the two success events (2) at radius $h/6$ are measurable. Then

$$\max_{i=0,1} \Pr_{\rho_i} \{L(\hat{\rho}, \rho_i) > h/6\} \geq \frac{1 - B_n}{2} > \frac{1}{4}, \quad B_n = \frac{1}{2} \sqrt{\exp(n\pi h^4/8) - 1}. \quad (5)$$

In particular, for $n \geq 16$, every estimator with measurable success events for all $\rho \in \mathcal{K}$ has, under some $\rho \in \mathcal{K}$, error strictly greater than $n^{-1/4}/6$ with probability strictly greater than $1/4$.

For randomized estimators, measurability of evaluation at each point alone need not imply measurability of an uncountable simultaneous success event. The explicit success-event assumption is therefore part of the theorem. Deterministic estimators on $\mathcal{O}_n$ satisfy it.

The preceding two-point result remains useful at smaller sample sizes. The next result closes the logarithmic gap in the same model. Set $a_n = (\log n/n)^{1/4}$.

Theorem 1.4 (Logarithmic randomized lower bound). *For every integer $n \geq 2^{64}$ and every estimator $b : \mathcal{O}_n \times \Omega \rightarrow \mathbb{R}^D$, let ξ be an arbitrary probability law for its independent seed. Assume the success events (2) at radius $a_n/8192$ are measurable for every $\rho \in \mathcal{K}$. Then some $\rho \in \mathcal{K}$ satisfies*

$$(\pi_n^\rho \otimes \xi)\{L(b, \rho) > a_n/8192\} \geq \frac{1}{2}. \quad (6)$$

The same conclusion holds at any smaller radius whose success events are measurable. No regularity or class membership of the outputs is required.

For a fixed seed law ξ , let $r_{95, \xi}(n)$ be the infimum of all nonnegative radii r for which one estimator has measurable success events at r and success probability at least 19/20 for every $\rho \in \mathcal{K}$. A one-point seed includes deterministic estimation.

Corollary 1.5 (Fixed-confidence minimax rate). *For every probability seed space and $n \geq 2^{64}$,*

$$\frac{a_n}{8192} \leq r_{95, \xi}(n) \leq \min\{1/2, 650a_n\}. \quad (7)$$

Consequently the same bounds hold if the infimum also ranges over independent randomized procedures.

This is rate optimality up to constants at fixed confidence. It makes no assertion about a sharp leading constant or expected risk. The lower proof applies to full coordinate observations, while the upper estimator uses order only. Both experiments therefore have the same rate for this loss; equal finite-sample information or equal optimal constants do not follow.

1.1 Relation to earlier work

The geometric input is the occupied-grid realizer theorem in the author’s reconstruction paper [1]. For completeness, Section 2 repeats its forcing argument and probability calibration. Coordinate-pattern forcing is earlier work: Winkler’s 1985 Theorem 7 and his dimension-two analysis [2, 3] are direct precedents. In the independent uniform-permutation model, Winkler’s Lemmas 2.2–2.6 and Theorem 4.3 give stronger typical rank rigidity, up to a global axis exchange. Nonconstant members of $\mathcal{K}$ do not give independent coordinate permutations, so that flat-model conclusion is not imported here.

Copula density estimation from coordinate data or paired empirical ranks is also established. Genest, Masiello and Tribouley [4] analyze wavelet estimators in integrated quadratic and pointwise expected loss. Autin, Le Pennec and Tribouley [5] study adaptive thresholding and quadratic risk. These are distinct observation and loss specifications from a finite-confidence, order-only supremum bound.

The published Seck–Mamane result [6], Theorem 1 (p. 40), concerns rank-based wavelets and almost-sure uniform convergence under wavelet and Besov conditions. Its smoothness condition is $(d+2)/2 < t < N+1$; with $2^{j_n} \asymp (n/\log n)^{1/(2t+d)}$, its displayed rate is

$$\left(\frac{2^{2(1+d)j_n} \log \log n}{n} \right)^{1/2}.$$

Remark 3 (p. 43) distinguishes this from the optimal oracle rate. The published theorem differs from the earlier 2023 preprint, so we use the published statement. Swanepoel, Seck and Mamane [7] obtain an almost-sure supremum rate via joint and marginal density estimates and quantiles,

with their own Besov, support and positive marginal assumptions. Neither almost-sure statement is converted here into a uniform finite 95% radius. Smoothness of each ρ does not supply uniform higher-regularity norms over $\mathcal{K}$.

Braun’s qualitative reconstruction results [8] concern all-size chronology laws under stated geometric hypotheses, with dimension at least three. The present fixed-gauge, two-dimensional statistical experiment does not extend those theorems or settle general spacetime reconstruction. The separate coordinate-gauge paper [9] exhibits a higher-dimensional obstruction to a proposed gauge; it is a different manuscript and result.

The candidate contribution here is the degree-filter/deep-cell reduction in Section 3, composed with geometric recovery to obtain (3), and its exact formal verification. The new contribution in this revision is a matching lower bound in the exact original class and quotient, and a certified minimax-radius sandwich. Histogram bias, Chernoff bounds, likelihood products and testing reductions are standard methods. In particular, the second-moment many-hypothesis inequality used below is the $l = 2$ case of the established power-divergence route in Guntuboyina [11], Example II.8, equation (15), following the finite-packing reduction in Section I and Theorem II.1. We write out the elementary inequality and its seed and measurability details; we do not claim a new general testing method. The bounded primary-source comparison found no inspected theorem with the full experiment, class, quotient and confidence specification of Theorem 1.1; this is a provisional priority assessment, not an exhaustive absence claim.

2 Geometric recovery on a finite-probability event

Fix $m \geq 16$, $r = 1/m$. Let E require every open grid cell of side r to be occupied, and the empirical anchored distribution function $C_n(s, t) = n^{-1} \#\{i : U_i \leq s, V_i \leq t\}$ to differ from its population version $C(s, t)$ by at most r at each grid vertex. We also intersect with the probability-one event of distinct coordinates in $(0, 1)$ and no coordinate on any grid line. The latter restrictions do not change any probability bounds.

Lemma 2.1 (Grid probability and interpolation). *Uniformly over $\rho \in \mathcal{K}$,*

$$\Pr_\rho(E^c) \leq m^2 e^{-nr^2/2} + 2(m+1)^2 e^{-2nr^2}. \quad (8)$$

On E the two marginal empirical errors at arbitrary thresholds in $[0, 1]$ are at most $2r$, the joint error is at most $3r$, and the fraction of points outside $J = [4r, 1 - 4r]^2$ is at most $20r$.

Proof. Every cell has probability at least $r^2/2$, so its empty probability is at most $(1 - r^2/2)^n \leq e^{-nr^2/2}$. At a fixed grid vertex, Hoeffding’s inequality [10] gives a two-tail probability at most $2e^{-2nr^2}$. A union bound proves (8). For clarity, the bounded-variable MGF behind this step is $\mathbb{E}e^{t(X - \mathbb{E}X)} \leq e^{t^2/8}$ for $0 \leq X \leq 1$: the second derivative of its log MGF is the variance under exponential tilting, at most $1/4$, and the value and first derivative at zero vanish. Independence, Markov’s inequality and $t = 4r$ yield one tail e^{-2nr^2} ; use $-t$ for the other.

At grid thresholds, each marginal is an anchored rectangle with the other threshold 1, so its error is at most r . Monotonicity and uniform marginals add at most r between adjacent grid thresholds. For the joint CDF, enlarge or shrink the rectangle to a grid vertex. The population difference is at most $2r$ because it lies in two marginal strips. Adding the vertex error gives $3r$. Each of the four marginal tails beyond $4r$ and $1 - 4r$ has empirical mass at most $5r$. Their union contains all points outside J . $\square$

A realizer is a pair of strict total orders whose intersection is $\prec$. Its inclusive rank is $\text{rank}_L(i) = 1 + \#\{j : L(j, i)\}$. Distinct latent coordinates supply a realizer, although neither latent total order is observed.

Lemma 2.2 (Earlier realizer rigidity). *On E , every realizer has one global exchange of its two orders after which, for every $Z_i \in J$,*

$$\left| \frac{\text{rank}_{L_1}(i)}{n} - U_i \right| \leq 32r, \quad \left| \frac{\text{rank}_{L_2}(i)}{n} - V_i \right| \leq 32r. \quad (9)$$

Proof. This is the finite theorem of [1]; we give its argument to specify the dependency. For incomparable x, y write $Q(x, y)$ when $L_1(x, y)$ and $L_2(y, x)$. The relation Q is transitive. If incomparability edges xy, xz share x and y, z are comparable, their orientations at x agree; otherwise a two-edge Q path would make y, z incomparable.

Occupancy provides

$$\begin{aligned} A &\in (r, 2r) \times (1 - 2r, 1 - r), \\ B &\in (1 - 2r, 1 - r) \times (r, 2r), \\ W &\in (1 - r, 1) \times (1 - 3r, 1 - 2r). \end{aligned}$$

For interior incomparable points x, y with $u_x < u_y$ and $v_x - v_y \geq 3r$, choose an open grid row inside (v_y, v_x) . Explicitly, $k = \lfloor v_y/r \rfloor + 1$ has $kr > v_y$ and $(k + 1)r \leq v_y + 2r < v_x$; interior membership makes this row admissible. The first cell in that row supplies z with $0 < u_z < r$. The incomparability edges

$$xy, \quad zy, \quad Ay, \quad AW, \quad AB$$

have comparable opposite endpoints at each step: $z \prec x$, $z \prec A$, $y \prec W$, $B \prec W$. Thus $Q(x, y)$ has the same truth value as $Q(A, B)$. Exchanging the two orders once makes $Q(A, B)$ agree with increasing latent u . All such separated interior incomparable pairs then agree with the latent realizer; comparable pairs already agree.

For a fixed interior point Z_i , a rank disagreement can therefore involve only a point outside J or one in $|V_j - V_i| < 3r$. The latter strip has empirical mass at most $6r + 4r = 10r$ by Lemma 2.1; both endpoints lie in $[0, 1]$. There are at most $30rn$ possible disagreements, bounding each rank difference by $30rn$. Finally, the inclusive true rank divided by n equals the marginal empirical CDF at the point and is within $2r$ of its coordinate. This proves (9). $\square$

The orientation can depend on the sample and realizer. It is a single choice for all retained points and all subsequent queries.

3 Observed degrees and the deep-cell reduction

Let $h = \sqrt{r}$ and retain precisely

$$I = \{i : d_i^- > 6rn, d_i^+ > 6rn\}, \quad (10)$$

where $d_i^- = \#\{j : j \prec i\}$ and $d_i^+ = \#\{j : i \prec j\}$. These counts use only the observed relation.

Lemma 3.1 (Two inclusions). *If E holds and $1/n \leq r$, then*

$$\{i : Z_i \in [7h, 1 - 7h]^2\} \subseteq I \subseteq \{i : Z_i \in J\}.$$

Proof. If $U_i < 4r$ or $V_i < 4r$, all predecessors are in the corresponding empirical lower marginal tail, of mass at most $5r$. If $U_i > 1 - 4r$ or $V_i > 1 - 4r$, all successors lie in an upper tail of mass at most $5r$. Such points fail (10).

For the converse use the exact strict-count identities

$$d_i^-/n = C_n(U_i, V_i) - 1/n, \quad (11)$$

$$d_i^+/n = 1 - F_{U,n}(U_i) - F_{V,n}(V_i) + C_n(U_i, V_i). \quad (12)$$

Coordinate injectivity makes the self-subtraction in (11) exact; inclusion-exclusion gives (12). At a deep point, southwest and northeast population rectangle masses are each at least $(7h)^2/2 = 49r/2$, by the lower density bound. The first empirical rectangle differs by at most $3r$, and the second by at most $2r + 2r + 3r = 7r$. Consequently

$$d_i^-/n \geq 49r/2 - 3r - 1/n \geq 41r/2 > 6r, \quad d_i^+/n \geq 49r/2 - 7r = 35r/2 > 6r.$$

No independence of degrees or retained points is used. $\square$

Now suppose $m \geq 65536$ and define

$$k = \lfloor \sqrt{m} \rfloor, \quad H = 8h, \quad s = 32r, \quad w = \frac{1 - 2H}{k}. \quad (13)$$

Partition the inner square into k^2 cells of side w , using intervals $(a, a + w]$ for counts. For a fixed realizer let $q_i = (\text{rank}_{L_1}(i)/n, \text{rank}_{L_2}(i)/n)$ and set

$$b_R = \frac{\#\{i \in I : q_i \in R\}}{nw^2}. \quad (14)$$

The denominator is the original n , not the number retained. For $R = (a, a + w] \times (c, c + w]$, write

$$R^- = (a + s, a + w - s] \times (c + s, c + w - s], \quad R^+ = (a - s, a + w + s] \times (c - s, c + w + s].$$

Fixed boundary lines are null for latent points. These half-open conventions also make the deterministic inclusions below valid at their relevant endpoints.

Lemma 3.2 (Mesh and cell sandwich). *The parameters (13) satisfy*

$$0 < h \leq \frac{1}{256}, \quad \frac{15}{16}h \leq w \leq 2h, \quad 2s \leq \frac{h}{4} < w, \quad H - s \geq \frac{63}{8}h > 7h. \quad (15)$$

If E holds and $1/n \leq r$, one $S \in G$ simultaneously gives, for every cell,

$$\frac{N(SR^-)}{n} \leq \frac{\#\{i \in I : q_i \in R\}}{n} \leq \frac{N(SR^+)}{n}, \quad (16)$$

where $N(A) = \#\{i : Z_i \in A\}$.

Proof. We have $\sqrt{m}/2 \leq k \leq \sqrt{m}$, $k^2 \leq m$ and $15/16 \leq 1 - 16h < 1$, giving the width inequalities. The other bounds follow from $s = 32h^2$ and $h \leq 1/256$. Every expanded cell, including its closure, lies in $[7h, 1 - 7h]^2$ by the final margin bound.

Fix the one alignment in Lemma 2.2; equivalently $|q_i - SZ_i|_\infty \leq s$ for every retained point. If $SZ_i \in R^-$, Lemma 3.1 retains i , and its coordinate error places q_i in R . If $q_i \in R$ and i is retained, the same error puts SZ_i in R^+ . These implications prove (16). Both squares and the degree filter are invariant under transpose, so the same S works throughout. $\square$

This is the key reduction: local true points are all retained. There is no global discarded-count term divided by the cell area.

4 Local concentration and full-square estimation

Lemma 4.1 (Small-mass concentration). *For independent Bernoulli variables of common mean $p \leq 8h^2$, where $0 < h \leq 4$, their average $\overline{X}$ satisfies*

$$\Pr\{|\overline{X} - p| > 4h^3\} \leq 2e^{-nh^4/2}. \quad (17)$$

Proof. For $|t| \leq 1$, the exponential series gives $e^t - 1 - t \leq t^2 \sum_{j \geq 2} 1/j! \leq t^2$. Here $e < 3$, as also shown in Section 5. Thus

$$\mathbb{E}e^{t(X-p)} = e^{-tp} \{1 + p(e^t - 1)\} \leq e^{p(e^t - 1 - t)} \leq e^{pt^2}.$$

Multiply these MGFs, apply Markov's inequality and choose $t = h/4$. The exponent for tolerance $4h^3$ is at most $-n(h/4)(4h^3) + n(8h^2)(h/4)^2 = -nh^4/2$. The same calculation with $-t$ proves the other tail. $\square$

The family of rectangles $SR^\pm$, over every cell and both $S \in G$, is fixed before sampling. Each member satisfies

$$Q_\rho(SR^\pm) \leq \frac{3}{2}(w + 2s)^2 \leq \frac{3}{2}(9h/4)^2 = \frac{243}{32}h^2 < 8h^2. \quad (18)$$

Counting the two signs and two orientations gives at most $4k^2$ rectangles. Let F be the event that all their empirical masses are within $\varepsilon = 4h^3$ of their true masses. Then

$$\Pr(F^c) \leq 8k^2 e^{-nh^4/2}. \quad (19)$$

Some rectangles duplicate one another under transpose; counting both orientations is the conservative bound used by the formal proof. We do not assume that recovered points are independent, or that E and F are independent.

Lemma 4.2 (Cell error and boundary extension). *With $m \geq 65536$ and $1/n \leq r$, on $E \cap F$, clip each b_R to $[1/2, 3/2]$. Clamp each query coordinate into $[H, 1 - H]$ and return the clipped value of a cell whose closure contains the clamped point. This gives, under the same S , a full-square error at most $270h$.*

Proof. Put $\rho^S = \rho \circ S$ and $p_R = \int_R \rho^S$. The expansion shell has area $4ws + 4s^2$, so its mass is at most $6ws + 6s^2$. The contraction loss is no larger. The count sandwich and event F give

$$|b_R - p_R/w^2| \leq 6s/w + 6(s/w)^2 + 4h^3/w^2. \quad (20)$$

At every point y in the closed cell, the difference between $\rho^S(y)$ and its cell average is at most $2w$. Indeed, for a uniform point X in the cell, $\mathbb{E}|X - y|_2^2 \leq 2w^2/3$, with the largest value at a corner; Cauchy-Schwarz and the Euclidean Lipschitz constant give $2\mathbb{E}|X - y|_2 \leq 2w$.

Clipping cannot increase error relative to a number in $[1/2, 3/2]$. Clamping moves a query point by at most $\sqrt{2}H$, so its density cost is at most $2\sqrt{2}H < 3H = 24h$. Using (15) in (20), we obtain respectively

$$6s/w \leq \frac{3072}{15}h, \quad 6(s/w)^2 \leq \frac{6144}{225}h, \quad 4h^3/w^2 \leq \frac{1024}{225}h.$$

The sum of these three constants is $53248/225$. Adding $2w \leq 4h$ and $24h$ gives $(59548/225)h < 270h$, uniformly on the closed square. $\square$

To fix the query convention exactly, set

$$j(x) = \min \left\{ \left\lfloor \frac{\text{clamp}_{[H, 1-H]}(x) - H}{w} \right\rfloor, k - 1 \right\}.$$

Use cell $(j(u), j(v))$. Internal query edges select the cell to their right, while count cells are open on the left and closed on the right. Lemma 4.2 holds on each cell closure, so this distinction does not affect any query, including corners. The coordinatewise convention commutes with transpose. Finite cell selection and clamping also prove query measurability.

4.1 The selector and every sample-size branch

For $n \geq 2$, let

$$x_n = \sqrt{\frac{n}{8 \log n}}, \quad m = \lfloor x_n \rfloor. \quad (21)$$

If $m < 65536$, return the flat function 1. Otherwise put $h = m^{-1/2}$. If $270h \geq 1/2$, again return 1. These decisions depend only on n . In the remaining branch use the construction above.

Choose once, on the finite space $\mathcal{O}_n$, one labeled representative of each isomorphism class and one realizer whenever one exists. Finite enumeration is one possible specification; the Lean selector uses finite classical choices. Return 1 if there is no realizer. On actual samples there is a realizer almost surely. Relabeling permutes the predecessor and successor sets and the rank counts, so the construction transports to the selected representative. The proofs hold for every realizer, with its own single global alignment. Thus the choice uses no unknown ρ , and the guarantee descends to the actual unlabeled observation. No efficient algorithm for making these choices is asserted.

Proof of Theorem 1.1. In the active branch $m \geq 65536$, and

$$nr^2 = n/m^2 \geq 8 \log n, \quad m^2 \leq n/8, \quad n \geq 4m^2, \quad 1/n \leq r.$$

Here $n \geq 4m^2$ follows first from $\log n \geq \log 2 > 1/2$; this implies $n \geq 4$ in this branch, hence $\log n > 1$ and $m^2 \leq n/8$. Since $(m+1)^2 \leq 4m^2$, Lemma 2.1 gives

$$\Pr(E^c) \leq \frac{1}{8n^3} + \frac{1}{n^{15}}. \quad (22)$$

Also $h^4 = r^2$, $k^2 \leq m \leq n$, so (19) gives $\Pr(F^c) \leq 8/n^3$. Already for $n \geq 8$,

$$\frac{1}{8n^3} + \frac{1}{n^{15}} + \frac{8}{n^3} < \frac{1}{20}.$$

Lemma 4.2 consequently supplies the $270h$ radius with probability at least $19/20$. This is the certified intermediate Chernoff budget, not the sharper Bernstein budget of an earlier working derivation.

For $m \geq 65536$, $\lfloor x_n \rfloor \geq x_n/2$, whence

$$270h \leq 270\sqrt{2} 8^{1/4} \left(\frac{\log n}{n} \right)^{1/4} < 650 \left(\frac{\log n}{n} \right)^{1/4}.$$

The numerical comparison is $32 \cdot 270^4 < 650^4$. In the active branch $270h < 1/2$ as well, giving the minimum in (3).

If $m \geq 65536$ but $270h \geq 1/2$, the same scale comparison gives $R_n = 1/2$. If $m < 65536$, then $x_n < 65536$ and

$$650 \left(\frac{\log n}{n} \right)^{1/4} > \frac{650}{8^{1/4} 256} > \frac{1}{2},$$

so again $R_n = 1/2$. In both fallback branches the flat function has error at most $1/2$ deterministically, by (1). This covers every $n \geq 2$ and completes the selector proof. $\square$

Proof of Corollary 1.2. Let A_ρ, A_σ be the two success sets of the same estimator at radius R_n , and put $\delta = \text{TV}(\pi_n^\rho, \pi_n^\sigma)$. Under π_n^σ , their intersection has probability at least $19/20 - \delta + 19/20 - 1 = 9/10 - \delta$. If $\delta < 9/10$, this intersection is nonempty. At any observation in it, the two global alignments and the triangle inequality imply $d_{\text{conf}}(\rho, \sigma) \leq 2R_n \leq 1300(\log n/n)^{1/4}$. If $\delta \geq 9/10$, the class bounds give $d_{\text{conf}}(\rho, \sigma) \leq 1 \leq (10/9)\delta$. In all cases $d_{\text{conf}} \leq 1$, yielding (4). $\square$

5 Smooth alternatives and testing

Lemma 5.1 (Original-class membership and separation). *For $0 < h \leq 1/2$, the two alternatives in Theorem 1.3 belong to $\mathcal{K}$, are invariant under transpose, and have $d_{\text{conf}}(\rho_0, \rho_1) = h/e > h/3$.*

Proof. The profile $f(t) = te^{-t^2}$ is smooth and odd, with

$$\sup |f| = \frac{1}{\sqrt{2e}} < \frac{1}{2}, \quad f'(t) = (1 - 2t^2)e^{-t^2}, \quad \sup |f'| \leq 1.$$

The first maximum occurs at $|t| = 1/\sqrt{2}$. For the derivative, on $t^2 \leq 1/2$ the bound is immediate; on $t^2 \geq 1/2$ the maximum of $(2t^2 - 1)e^{-t^2}$ is $2e^{-3/2} < 1$, at $t^2 = 3/2$. Since $a_h(1 - x) = -a_h(x)$, its integral on $[0, 1]$ vanishes exactly. Consequently both marginal integrals of ρ_1 equal one. Also $|\rho_1 - 1| \leq h/2 \leq 1/4$, and each partial derivative has magnitude at most $2 \sup |f'| \sup |f| \leq 1$. The Euclidean gradient bound $\sqrt{2} < 2$, integrated along segments in D , proves (1). Smoothness holds on all of $\mathbb{R}^2$.

Both densities are transpose invariant. At $u = v = 1/2 + h/\sqrt{2}$, which lies in D , the deviation is h/e . The profile maximum bounds the deviation everywhere by the same number, proving exact quotient separation. Finally $e < 3$: for $j \geq 2$, $j! \geq 2^{j-1}$, with strict inequality for $j \geq 3$, so $\sum_{j \geq 2} 1/j! < 1$. Thus $h/e > h/3$. $\square$

Lemma 5.2 (Likelihood product and order data processing). *For the alternatives above and every $n \geq 1$,*

$$\text{TV}(\pi_n^{\rho_0}, \pi_n^{\rho_1}) \leq \frac{1}{2} \sqrt{\exp(n\pi h^4/8) - 1}.$$

Proof. With Q_0 uniform on D ,

$$c := \chi^2(Q_1, Q_0) = \int_D (\rho_1 - 1)^2 = 4h^2 \left(\int_0^1 a_h(x)^2 dx \right)^2 \leq \frac{\pi}{8} h^4.$$

To check the constant, substitute $t = (x - 1/2)/h$ and enlarge the nonnegative integral to the real line:

$$\int_{\mathbb{R}} t^2 e^{-2t^2} dt = \frac{1}{4} \sqrt{\frac{\pi}{2}}, \quad \left(\int_{\mathbb{R}} t^2 e^{-2t^2} dt \right)^2 = \frac{\pi}{32}.$$

Indeed, squaring the Gaussian integral and using polar coordinates gives $\int_{\mathbb{R}} e^{-2t^2} dt = \sqrt{\pi/2}$. Integrating the derivative of te^{-2t^2} on expanding symmetric intervals, whose boundary term tends to zero, gives the factor $1/4$.

The actual likelihood ratio of $Q_1^{\otimes n}$ against $Q_0^{\otimes n}$ is $W(z_1, \dots, z_n) = \prod_i \rho_1(z_i)$. Fubini, $\int \rho_1 = 1$ and $\int \rho_1^2 = 1 + c$ give

$$\int W = 1, \quad \int W^2 = (1 + c)^n, \quad \int (W - 1)^2 = (1 + c)^n - 1 \leq e^{n\pi h^4/8} - 1.$$

Therefore Cauchy–Schwarz gives $\text{TV}(Q_0^{\otimes n}, Q_1^{\otimes n}) = \frac{1}{2} \int |W - 1| \leq \frac{1}{2} \sqrt{\int (W - 1)^2}$. The sampled directed relation is a finite collection of strict Borel inequalities; passing to an isomorphism class is a finite map. Pulling back an order event proves that its probability difference cannot exceed coordinate-data TV. This proves the claim for the actual unlabeled order laws. $\square$

Proof of Theorem 1.3. Set $r_* = h/6$. By Lemma 5.1, the closed success sets at radius r_* for ρ_0, ρ_1 are disjoint: otherwise the triangle inequality after composing their global alignments would give $d_{\text{conf}}(\rho_0, \rho_1) \leq 2r_* = h/3$. This argument applies to arbitrary outputs; it uses only the two all-point success inequalities.

Write $P_i = \pi_n^{\rho_i}$. If a seed has probability law ξ , independent of the observation under either hypothesis, every measurable event $A \subseteq \mathcal{O}_n \times \Omega$ has

$$|(P_0 \otimes \xi)(A) - (P_1 \otimes \xi)(A)| \leq \int |P_0(A_\omega) - P_1(A_\omega)| d\xi(\omega) \leq \text{TV}(P_0, P_1).$$

Sections are measurable and the sum over the finite observation space justifies the integral without additional output regularity. Decide hypothesis zero exactly on its success event A . Under zero the test error is strict estimation failure; under one, disjointness makes test error a subset of strict estimation failure. The sum of these two risks is at least

$$(P_0 \otimes \xi)(A^c) + (P_1 \otimes \xi)(A) = 1 - \{(P_0 \otimes \xi)(A) - (P_1 \otimes \xi)(A)\} \geq 1 - B_n.$$

Their maximum is therefore at least $(1 - B_n)/2$.

For the selected h , $nh^4 \leq 1$: if $n \leq 16$ use $h = 1/2$; if $n \geq 16$ use $h = n^{-1/4}$. Since $\pi < 4$ and $e^{1/2} < 2$, we have $B_n < 1/2$. The latter exponential inequality also follows from $e < 3 < 4$. Thus the finite lower bound is strictly greater than $1/4$. Both alternatives are in $\mathcal{K}$, and for $n \geq 16$ the radius is exactly $n^{-1/4}/6$, proving the final assertion. $\square$

5.1 What follows for confidence bands

Suppose measurable simultaneous bands $[\ell(x), u(x)]$ have $\ell \leq u$, with coverage meaning containment of $\rho \circ S$ at every $x \in D$ for one $S \in G$. Assume the relevant band and midpoint success events are measurable. If the maximum width $W = \sup_D (u - \ell)$ is at most $2r_*$ on every output, coverage implies midpoint loss at most r_* . Such a band cannot have uniform 95% coverage at this width. If the width is random, the valid implication is

$$\max_{i=0,1} \left[\Pr_{\rho_i} \{\text{coverage fails}\} + \Pr_{\rho_i} \{W > 2r_*\} \right] \geq \frac{1 - B_n}{2}. \quad (23)$$

Indeed midpoint failure is contained in the union of these two events, and Theorem 1.3 applies to the midpoint. This does not exclude occasional narrow reports or assert anything about coverage conditional on accepting a report. Equation (23) is a written consequence, not a separately exported Lean band theorem.

6 A matching logarithmic obstruction

We retain $f(t) = te^{-t^2}$ from Section 5, but translate and center it to allow a growing family of alternatives.

Lemma 6.1 (Centered profiles). *For $h > 0$ and $c \in \mathbb{R}$, define*

$$g_{h,c}(x) = f((x-c)/h), \quad \mu_{h,c} = \int_0^1 g_{h,c}(x) dx, \quad p_{h,c}(x) = g_{h,c}(x) - \mu_{h,c}.$$

The profile is smooth on $\mathbb{R}$, has zero integral on $[0, 1]$, and

$$\begin{aligned} |\mu_{h,c}| &\leq h/2, & |p_{h,c}(x)| &\leq 1/2 + h/2, \\ |p_{h,c}(x) - p_{h,c}(y)| &\leq |x - y|/h, & \int_0^1 p_{h,c}^2 &\leq h. \end{aligned} \tag{24}$$

If $h \leq 1/16$, then $p_{h,c}(c + h/2) \geq 1/4$ and $|p_{h,c}(x)| \leq 1/8$ whenever $|(x - c)/h| \geq 4$.

Proof. An antiderivative of $g_{h,c}$ is $-(h/2)e^{-((x-c)/h)^2}$, so

$$\mu_{h,c} = \frac{h}{2}(e^{-(c/h)^2} - e^{-((1-c)/h)^2}).$$

Both exponentials lie in $[0, 1]$, proving the mean bound. The bounds $|f| \leq 1/2$ and $|f'| \leq 1$ proved earlier give the absolute and Lipschitz bounds. Centering gives exact zero mean and

$$\int_0^1 p_{h,c}^2 = \int_0^1 g_{h,c}^2 - \mu_{h,c}^2 \leq h \int_{\mathbb{R}} f(t)^2 dt = \frac{h\sqrt{\pi}}{4\sqrt{2}} \leq h.$$

All interval integrands are continuous; the whole-line integral is the finite Gaussian moment evaluated in Lemma 5.2. Since $e^{-1/4} \geq 3/4$, $f(1/2) \geq 3/8$; subtracting a mean of absolute value at most $h/2$ proves the peak bound. For $|t| \geq 4$,

$$e^{t^2} \geq (1 + t^2/2)^2 \geq t^4/4 \geq 16|t|,$$

using $e^s \geq 1 + s$ and $|t|^3 \geq 64$. Thus $|f(t)| \leq 1/16$, and centering gives $|p_{h,c}(x)| \leq 1/16 + h/2 \leq 3/32 \leq 1/8$ at the stated tail points. $\square$

Lemma 6.2 (An admissible finite packing). *For an integer $m \geq 1$, put*

$$h = \frac{1}{16m}, \quad c_j = \frac{1}{4} + \frac{j+1/2}{2m}, \quad \rho_j(u, v) = 1 + \frac{h}{2}p_{h,c_j}(u)p_{h,c_j}(v), \quad 0 \leq j < m. \tag{25}$$

All ρ_j belong to $\mathcal{K}$ and are invariant under transpose. For $j \neq k$ they have quotient separation at least $3h/128$. Consequently their success sets at any radius $r \leq h/256$ are pairwise disjoint, even for arbitrary estimator outputs.

Proof. We have $0 < h \leq 1/16$ and $|p_{h,c_j}| \leq 1$. Smoothness holds globally. Zero profile mean makes each marginal integral exactly one, and $|\rho_j - 1| \leq h/2 \leq 1/32$ implies the required range. Expanding a product difference and applying Lemma 6.1 gives, for $z, w \in D$,

$$|\rho_j(z) - \rho_j(w)| \leq \frac{|z_1 - w_1| + |z_2 - w_2|}{2} \leq 2|z - w|_2.$$

This verifies the original Euclidean Lipschitz condition, without a uniform higher-derivative assumption.

The centers lie in $[1/4, 3/4]$ and satisfy $|c_j - c_k| \geq 8h$. The point $x_j = c_j + h/2$ lies in $[0, 1]$. At x_j the own profile is at least $1/4$, whereas every other profile has absolute value at most $1/8$, since $|(x_j - c_k)/h| \geq 8 - 1/2 \geq 4$. At the diagonal witness $z_j = (x_j, x_j)$ it follows that

$$\rho_j(z_j) - \rho_k(z_j) \geq \frac{h}{2} \left(\frac{1}{16} - \frac{1}{64} \right) = \frac{3h}{128}. \quad (26)$$

Transpose fixes each alternative, so taking the quotient does not decrease their supremum separation. Two successes would, at z_j , give a difference at most $2r \leq h/128$, contradicting (26). This proof uses only pointwise inequalities. $\square$

Lemma 6.3 (Product and independent-seed moments). *Let Q_0 be uniform measure on D , and $Q = Q_0^{\otimes n} \otimes \xi$. The likelihood of $Q_{\rho_j}^{\otimes n} \otimes \xi$ against Q is $L_j(z, \omega) = \prod_{i=1}^n \rho_j(z_i)$, and*

$$\int L_j dQ = 1, \quad \int L_j^2 dQ \leq B := e^{nh^4/4}. \quad (27)$$

Proof. Zero marginal means cancel the linear term in ρ_j^2 . Fubini and Lemma 6.1 give

$$\int_D (\rho_j - 1)^2 = \frac{h^2}{4} \left(\int_0^1 p_{h,c_j}^2 \right)^2 \leq \frac{h^4}{4}, \quad \int_D \rho_j^2 \leq 1 + h^4/4.$$

The densities are positive and bounded on D , so product likelihoods and their squares are integrable. Independence gives $\int L_j^2 \leq (1 + h^4/4)^n \leq e^{nh^4/4}$ and $\int L_j = 1$. Integration over an independent probability seed contributes the factor one. In particular, for every measurable event A the actual sample-times-seed probability is $\int_A L_j dQ$. $\square$

Lemma 6.4 (Disjoint-event testing). *Let Q be a probability measure and $P_0, \dots, P_{m-1}$ probability measures with integrable likelihoods L_j and $\int L_j^2 dQ \leq B$. For pairwise disjoint measurable sets A_j ,*

$$\sum_{j=0}^{m-1} P_j(A_j) \leq \sqrt{mB}. \quad (28)$$

If $m \geq 1$ and $B \leq m/4$, some $P_j(A_j^c) \geq 1/2$.

Proof. Set $F = \sum_j 1_{A_j} L_j$. Disjointness implies $F^2 = \sum_j 1_{A_j} L_j^2$ pointwise. Each summand is integrable, and Cauchy–Schwarz on the probability space gives

$$\sum_j P_j(A_j) = \int F dQ \leq \int |F| dQ \leq \sqrt{\int F^2 dQ} \leq \sqrt{\sum_j \int L_j^2 dQ} \leq \sqrt{mB}.$$

If $B \leq m/4$, average success is at most $1/2$. Taking complements under the probability measures proves the final assertion. $\square$

Lemma 6.5 (Explicit integer scale). *For $n \geq 2^{64}$ put $t = \log n$, $x = (n/t)^{1/4}$, $m = \lceil x \rceil$ and $h = 1/(16m)$. Then $m \geq 16$, $x \leq m \leq 2x$, $e^{nh^4/4} \leq m/4$, and $a_n/8192 \leq h/256$.*

Proof. The inequality $\log 2 \geq 1/2$ gives $t \geq 32$. For $t \geq 0$, $e^{t/2} \geq (1 + t/4)^2 \geq t$, where the last difference equals $(t - 4)^2/16$. Taking logarithms yields $\log t \leq t/2$, hence $\log x = (t - \log t)/4 \geq t/8 \geq 4$. Since $e \geq 2$, $x \geq e^4 \geq 16$. The ceiling therefore satisfies $x \leq m < x + 1 \leq 2x$, and $n/m^4 \leq t$. Thus

$$\frac{nh^4}{4} = \frac{n/m^4}{262144} \leq \frac{t}{16} \leq \frac{\log m}{2}.$$

Exponentiating gives $B \leq \sqrt{m} \leq m/4$ since $m \geq 16$. Finally $a_n = 1/x$ and $h/256 = 1/(4096m) \geq 1/(8192x) = a_n/8192$. $\square$

Proof of Theorem 1.4. Choose the family and scale of Lemmas 6.2 and 6.5. Pull back each success event at the target radius through the map from coordinates and seed to unlabeled order and seed. The map is measurable, since it consists of finitely many strict coordinate comparisons and a finite quotient map, with the identity on the seed. The pullbacks A_j are therefore measurable and pairwise disjoint. Their actual probabilities are the likelihood integrals in Lemma 6.3. Lemma 6.4 and $B \leq m/4$ give an alternative whose complementary event has probability at least $1/2$. The complement is strict failure of (2), including for unbounded outputs. Applying the argument directly at the requested radius uses no measurability assumption at an auxiliary radius. The same argument works at any smaller measurable radius. It also works for a coordinate-data estimator by using its success sets directly, without the order-map pullback. $\square$

Proof of Corollary 1.5. For a fixed seed space, Theorem 1.1 supplies an estimator that ignores its seed, so its success probability and measurability are unchanged. The set defining $r_{95,\xi}(n)$ is nonempty and bounded below by zero. Any achievable radius at most $a_n/8192$ would contradict Theorem 1.4: failure would be at least $1/2$ for some density but at most $1/20$ for all densities. Thus every achievable radius exceeds $a_n/8192$, and the infimum is at least this number. The upper bound follows from the achievable radius $\min\{1/2, 650a_n\}$. Both constants are independent of the seed law, which also permits the infimum over randomized procedures. $\square$

7 Formal verification and reproducibility

All main endpoints have exact-source remote GCP acceptance. Declarations below live in namespace `QuantyraNullCone`; a prefix `InDensityClass` indicates its namespace within that root. The companion `formal-map.json` supplies full identifiers, source paths and the retained exact-type/axiom evidence.

Upper theorem. `fourth_root_order_only_estimator` and `InDensityClass.fourth_root_density_estimation` in `DegreeEstimator.lean` prove Theorem 1.1 and its concrete selector instance. The estimator is quantified before the unknown density, and no good-event hypothesis remains in the endpoint.

Geometric and local reduction. `Cumulative.lean` supplies earlier coordinate rigidity. `DegreeFilter.lean`, `DegreeMass.lean` and `DegreeCells.lean` prove the strict-count identities, deep retention and one-orientation count sandwich. `BernoulliMGF.lean`, `SmallMassConcentration.lean` and `DegreeProbability.lean` supply the actual iid concentration.

Histogram and all- n assembly. `DegreeCellBias.lean`, `InnerHistogram.lean`, `DegreeHistogram.lean` and `DegreeMesh.lean` prove shell bounds, query coverage, clipping, clamping and the $270h$ constant. `DegreeHistogramProbability.lean`, `DegreeRateNumbers.lean` and `DegreeRelabel.lean` discharge mesh probabilities, flat branches and relabeling.

Actual-law corollary. `InDensityClass.fourth_root_actual_law_inverse` in `DegreeLaw.lean` proves Corollary 1.2.

Lower theorem. `fourth_root_randomized_lower_bound`, `fourth_root_lower_exists` and `fourth_root_minimax_obstruction` in `LowerBound.lean` give the finite two-alternative inequality, the existential full-class form and the $n \geq 16$ consequence. Dependencies are `OddGaussian.lean`, `LowerAlternatives.lean`, `LowerSeparation.lean`, `LowerMoments.lean`, `LowerTesting.lean`, `LowerScale.lean`, `LowerProduct.lean` and `LowerDataProcessing.lean`. The success-event hypotheses and independent seed are explicit.

Logarithmic lower theorem and minimax radius. `logarithmic_randomized_minimax_obstruction` and `logarithmic_lower_bound_at_radius` in `LogLowerBound.lean` prove Theorem 1.4 and its smaller-radius form. `logarithmic_minimax_radius_bounds` proves Corollary 1.5 for each probability seed space, using the formal infimum `orderConfidenceRadius95`. `LogLowerProfile.lean`, `LogLowerMoments.lean`, `LogLowerAlternatives.lean` and `LogLowerPacking.lean` discharge centered profiles, class membership and separation. `LogLowerTesting.lean`, `LogLowerProduct.lean` and `LogLowerScale.lean` discharge testing, iid and seed moments, and the explicit integer scale. No packing, moment, scale or good-event premise remains in the final endpoints. The coordinate-data and all-seed infimum consequences are written consequences of these arguments; they are not separately named Lean exports.

The proof/evidence revision is `a0b1f6846d6f0b89a94704ed2954645bcac756fd`. Lean 4.30.0 and mathlib revision `c5ea00351c28e24afc9f0f84379aa41082b1188f` are pinned. Authoritative GCP run `space-log-lower-final-acceptance-20261009T171053Z-0c7e20` completed 3035 root-build jobs and 508 selected exact-type/axiom audits with zero warnings. These counts cover the library, not 508 results asserted by this paper. All 143 captured proof/check/config identities and pinned dependencies were verified before and after execution. The audited axioms are only `propext`, `Classical.choice` and `Quot.sound`. Immutable inputs, failed attempts and raw diagnostics remain in the repository’s `evidence/gcp` tree; its `s036-logarithmic-final.json` indexes final acceptance. Every Lean/Lake invocation ran on GCP.

Manuscript preparation preserves those accepted bytes and invokes no new Lean build. The companion `verify_sources.py` checks the current and cited-commit hashes against the retained capture and audits; this byte check does not replace the manual statement-to-prose review. `build.py` compiles this source with Tectonic 0.17.0 and renders every PDF page with PyMuPDF for inspection. The package records the PDF hash, page review and compiler diagnostics. Formal checking establishes the encoded statements, not mathematical priority, model adequacy or practical efficiency.

8 Interpretation, practical limits and provenance

The result separates existence of a uniformly valid procedure from useful finite-sample performance. The chosen selector is active only when $270/\sqrt{m} < 1/2$, requiring $m > 291600$ with $m = \lfloor \sqrt{n}/(8 \log n) \rfloor$. The displayed constant is therefore deliberately conservative. It offers no practical density resolution at the tested sample sizes.

The separately frozen feasibility study [12] examined ten orders across five models at $n = 512$ and 3072 , producing thirty validated report envelopes over three reporting methods. Every density band had full width one. Twenty exact LP witness pairs documented the corner limitation.

Forty preparation/method runs stayed within the frozen 600-second and 4-GiB caps; the eighty conditional confirmation samples were not generated. The degree-filter estimator of this paper was not implemented or piloted as an additional method. Those results support neither practical success of this construction nor a contradiction of its asymptotic guarantee.

At $n = 3072$, the lower forbidden radius is approximately 0.0224, far below the study’s practical target radius 0.45. It cannot explain full-width pilot bands or show that the practical target is impossible. Better constants, efficient observable reconstruction and improved boundary calibration remain separate research tasks. The new multi-alternative theorem closes the rate gap, but its explicit threshold 2^{64} and small lower constant were chosen for a transparent proof, not practical sharpness. The upper constant 650 and the frozen pilot results are unchanged. No additional order-information penalty or higher-dimensional reconstruction theorem is proved.

This is a third, separate manuscript family in the shared proof repository. The reconstruction paper [1] addresses inverse laws and proper time; the gauge paper [9] addresses an explicit higher-dimensional coordinate obstruction. Their published artifacts and the frozen study are preserved. This is an unpublished 0.2.0 working revision of the finite-data manuscript. Its published 0.1.0 baseline remains at <https://doi.org/10.5281/zenodo.23265067>; that DOI does not identify this revised PDF. No publication of the revision is asserted here.

The work was developed with OpenAI Codex assistance under Daniel Eric Fredriksen’s research direction. AI-assisted derivation, formalization and writing are disclosed. Review follows an open-source, decentralized and informal workflow: inspectable proofs, reproducible evidence, downstream use, tests and versioned corrections. Specialist review and adoption are not prerequisites. Actual feedback must be distinguished from anticipated uptake. This manuscript and its companion artifacts are CC-BY-4.0; the proof library is Apache-2.0.

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